Joe Chang

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Passionate about geospatial technology, I spent three years at Google Maps honing my skills in ArcGIS, data management, and project leadership. Now, I'm eager to take the next step of combining my expertise with Python and application development to create more efficient mapping solutions that make a real impact.

GIS SKILLS

- ArcMap/Pro/Online
- ArcGIS/QGIS
- Project/DB Management
- Python/ArcPy/GeoPandas/SQL
- Arc FM/Survey123
- CAD/As-Built/Sketchup
- Georeferencing/Rectifying Raster

CODING SKILLS

- Web GIS Development
- GIS API
- React/JavaScript/Python
- MongoDB/mongoose
- UI/UX Design
- Testing and Debugging
- Source Control Management

EXPERIENCE

GIS Analyst — August 2024 - Present

Apple
Austin, TX (Remote)

- Implement quarterly updates for global users, delivering accurate addresses, enhanced automated search queries, and precise geolocation data for both new and existing features
- Develop and execute Python scripts to analyze nationwide metrics, optimizing strategies for implementing systemic improvements
- Perform comprehensive analyses to identify and resolve anomalies, investigating root causes of regressed performance to maintain system reliability

GIS Analyst/Project Lead — November 2020 - August 2023

Google Maps
Bothell, WA

- Imported over 100+ Low Emission Zone features into Google Maps for major cities across EU Nations, including London, Paris, and Rome
- Piloted project on Truck GPS covering over 1 million roads across the U.S. and Canada to ensure safe destination routing
- Processed open source datasets into automation-ready formats using ArcGIS.
- Improved data accuracy through validation, spatial assessments, and integrity checks, raising client satisfaction by 89%
- Documented technical procedures for the Team's workflow to standardize and improve the completion time, effectively improved TAT 140% from Q1 2023 – Q3 2023

PROJECTS

Low Emission Zones: Deployed over 100 Low Emission Zones across major European cities to restrict high-emission vehicles and reduce air pollution. Each zone specifies critical information such as geometric boundaries, restricted vehicles, exceptions, and links to official government sources. Information was provided in the respective country's official languages to ensure accessibility and precision.

Capstone Project: Collaboration with Forterra using census tract data in Snohomish County, WA to examine socio-ethnic factors linked to food deserts. Leveraged Enterprise GDB for simultaneous teamwork and data management, delivering a deployed Web Map and Accessible Geodatabase for Forterra.

NBA - App: Developed an interactive React-based application utilizing react-router-dom for navigation and Bootstrap for sytling. Integrated APIs to provide updated NBA news and league-wide data, as well as enabling User's to select specific teams to view specific stats with useState.

EDUCATION

Graduate Certificate in Software Design & Development Present
University of Washington

Post-Baccalaureate program. Focused on Object-Oriented Programming in C++, data structure/algorithms, systems programming, and software design/testing. Bridge pathway into Master of Science in Computer Science & Software Engineering

Full Stack Developer Boot Camp November 2023
Per Scholas

Comprehensive 16-week Full-Stack Software Engineering course that dives deep into the fundamental aspects of front and back-end software engineering: MERN stack, UI design, and System Architecture, and Back-End Development.

GIS Certificate June 2021
University of Washington

Capstone project in collaboration with Forterra to create an interactive GIS Web Map, visualizing areas of food disparities across Snohomish County, WA and capsulating a written analysis report on the correlations based on census tract data.

Environmental Studies B.A. minor: Geography June 2020
Western Washington University

Focused on Urban Planning with an emphasis on ethics, community engagement, and sustainable development. Completed a 10-week urban planning simulation project involving community engagement and development of a mock Comprehensive plan for Ferndale, WA.